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SECTION 22 05 23

GENERAL DUTY VALVES FOR PLUMBING

PART 1 GENERAL

1.1 General Requirements

- 1.1.1 Valves shall be installed only in vertical or horizontal positions unless otherwise required by the Drawings.
- 1.1.2 All valves shall be installed in accessible locations to facilitate easy removal for maintenance.
- 1.1.3 Valves shall be full-line size.
- 1.1.4 Valves 2" (50mm) dia. and smaller shall have threaded ends, valves 2½" (65mm) dia. and larger shall have flanged ends.
- 1.1.5 All threaded end valves shall be installed with unions to facilitate the removal of the valve from the pipeline.
- 1.1.6 Gate valves shall be installed on both sides of every piece of equipment for all pipe-system connections, and where shown on the Drawings.
- 1.1.7 All valves 2½" (65mm) and larger shall have internal epoxy coating suitable for drinking water application.

1.2 Related Works Specified Elsewhere

- 1.2.1 The works specified in the following divisions, sections and sub-sections are included in this Section in each applicable part, as if repeated herein verbatim.

Section 22 05 00 - General Mechanical Requirements for Plumbing

Section 22 07 00 - Plumbing Insulation

Section 22 11 13 - Facility Water Distribution Piping

Section 22 11 16 - Domestic Water Piping and Fittings

Section 22 11 23 - Domestic Water Pumps

Section 22 12 00 - Facility Potable Water Storage Tanks

Section 22 13 00 - Facility Sanitary Sewerage

Section 22 13 16 - Sanitary Waste and Vent Piping

Section 22 14 00 - Facility Storm Drainage

Section 22 33 30 - Domestic Water Heater

Section 22 40 00 - Sanitary Fixtures

1.3 Valves Pressure Rating

- 1.3.1 Unless specified otherwise all valves, strainers, flexible connections, etc. shall be selected for pressure rating at least 1.5 times the operating pressure.

- 1.3.2 Contractor shall submit a schedule of all valves, strainers, etc., showing the required pressure rating for each fitting and shall indicate its location and service.

1.4 Quality Assurance and Control

- 1.4.1 All valves shall be kite marked to the applicable B.S. standard.
- 1.4.2 All valves shall be full line size.
- 1.4.3 All valves in steam lines shall be suitable for steam application.

PART 2 PRODUCTS

2.1 Gate Valves

- 2.1.1 Gate valves shall be of non rising stem. Valves 2" (50 mm) and smaller, shall be all bronze, screwed ends and bronze wedge disc faces and seats. Valves 2½" (65 mm) and larger, shall be cast iron or ductile iron, flanged ends with, bolted bonnet and wedge disc faces and seats. Valves design and shall be leak free and shall conform to AWWA C515.
- 2.1.2 One gate valve shall be supplied and installed:
- a- At the supply and return from each equipment.
 - b- At the discharge and suction of each pump.
 - c- In general at all points shown on the Drawings and/or specified.

2.2 Check Valves (Spring Loaded Type)

- 2.2.1 Check valves shall be of the non-slam spring loaded, globe type with the capability to absorb and dampen the shock wave from the piping system water hammer. Check valves 2" (50 mm) and smaller, shall be all bronze, screwed ends. Valves 2½" (65 mm) and larger, shall be cast iron, flanged ends.
- 2.2.2 One check valve shall be installed:
- a- At the discharge of each pump.
 - b- At all points shown on the Drawings and/or where specified.

2.3 Float Valves

- 2.3.1 Float valves 2" (50 mm) and smaller shall be all bronze, screwed ends, float operated. Valves 2 ½" (65 mm) and larger shall be cast iron body, flanged ends, float operated. Float shall be all copper and mounted at the end of a brass or copper rod, which actuates valve operation.

2.4 Automatic Air Vents

- 2.4.1 Supply and install all automatic air vents as shown on the Drawings and wherever specified in this book of specifications.
- 2.4.2 Automatic air vents of the spherical float type shall be installed at all high points in the piping. Vents shall have cast iron housing and bolted cover with gasket. Float shall be constructed of stainless steel. Vents shall be suitable for a maximum operating pressure of 150 psi (1030 Kpa). A ½" (15 mm) lock shield valve shall be directly installed ahead of

each automatic air vent, and a ½" (15 mm) drain line shall be provided to discharge at a convenient point.

2.5 Backflow Preventers

2.5.1 This type of valve shall be used on domestic water systems, wherever applicable.

2.5.2 Valve shall combine the double check valve protection effect together with an air gap venting to the atmosphere, working on the reduced pressure principle.

2.5.3 Valve shall be suitable for 125 psi (860 Kpa) steam and 200 psi (1380 Kpa).

2.6 Pipe Expansion Joints

2.6.1 Expansion joints shall be provided wherever pipes cross structural expansion joints and wherever required to prevent undue stresses caused by thermal expansion of the pipes.

2.6.2 Expansion joints shall be of the packless-bellow type with flanged or welded ends as suitable for the pipe application.

2.6.3 Bellows shall be of stainless steel and suitable for a pressure of 125 psi (860 Kpa) or the design working pressure, whichever is greater. Expansion joints shall be provided with guides to prevent any unnecessary misalignment of the pipe. Guides and anchor arrangements shall be per the recommendations of the expansion joints manufacturers.

2.7 Balancing Valves

2.7.1 Supply and install balancing valves of double regulating effect with provision for connecting a portable differential pressure meter. Each meter connection shall have pressure temperature read out ports. Valves shall be of the Y-pattern style design which can be installed in any direction without affecting flow measurements.

2.7.2 All valves upto 2" (50 mm) shall be of bronze alloy. Valves of 2 ½" (65 mm) and above shall have cast iron body with all other metal parts of nonferrous copper alloy.

2.7.3 Each valve shall provide precise flow measurement, precision flow balancing and positive shut-off with no drip seat and teflon disc.

2.7.4 Each valve shall have multi-adjustment turns of hand wheel for maximum setting with hidden memory feature to program the valve with precision tamper proof balancing setting. Number of turns shall be 4 for valve up to 2" (50 mm), 8 between 2 ½" (65 mm) and 6" (150 mm), 12 for 8" (200 mm) and 10" (250 mm) and 16 for 12" (300 mm).

2.7.5 All valves up to 2" (50 mm) shall be supplied with polyurethane container.

2.7.6 One set of computerized balancing instrument shall be supplied with the valves. The instruments shall be a read out meter, programmed with the valve curves and with a conversion formula for Cv to read the flow in gpm directly. The instrument shall also be provided with temperature measurement probe.

2.7.7 A chucking device shall hold the valve disconnects firmly and a built-in flat seal shall give it maximum tightness.

2.7.8 The valves shall be globe style design and all metal parts of non-ferrous pressure die-cast, non porous metal: bronze alloy.

2.7.9 The valve shall be suitable for installing in any direction without affecting flow measurement and shall provide four functions:

- Precise flow measurement.
- Precision flow balancing
- Positive shut off with no drip seat and teflon disc.
- Drain connection with protective cap.

2.7.10 The valve shall have four 360 deg. adjustment turns of the hand wheel for maximum vernier type setting with hidden memory feature to program the valve with precision tamper-proof balancing setting.

2.8 Strainers

2.8.1 Water strainer shall be supplied and installed at the suction connection of all pumps, and ahead of all automatic flow control valves.

2.8.2 Strainers 2 1/2" (65mm) and smaller shall be Y-pattern type with bronze body, screwed cover, brass basket and screwed ends.

2.8.3 Strainers 3" (75 mm) and larger shall be of cast iron body, brass basket and flanged ends.

2.8.4 Basket shall have 1/32" (0.8 mm.) perforations for water service.

2.9 Flexible Connectors

2.9.1 Flexible Connectors shall be, stainless steel, easy flexing, long cyclic life connectors, to protect mechanical equipment by relieving piping stresses, caused by piping misalignment, sagging pipe hangers, and thermal expansion.

2.10 Accessories (Free Fall Device)

2.10.1 The valve shall be weight-operated with mercury switch devices opens on fusible link failure and shuts off the fuel oil supply pumps and the equipment is feeded by fuel oil such as boilers, incinerators, generators, etc., and so arranged that the weight falls freely, doing no work during the first part of its travel. By the time the weight is called upon to move the valve, it has accumulated sufficient momentum to overcome the inertia of the valve plug. Where pipe run is at low level special linkage should be supplied to raise lower position of dead weight to pipe centre line.

2.10.2 Required length of easy to fit non-corrosive, carefully selected stranded stainless steel cable to hold the valve open or closed as required.

2.10.3 Three brass screw hooks with brass pulleys which automatically align themselves to suit the run of the cable. Two of these shall be woodscrew thread, and one 6mm (–") B.S.P. thread for fitting to metal surfaces.

2.10.4 Copper tube cable connectors. Where the cable has to be cut and a loop formed, these connectors shall be quick and effective.

2.10.5 One fusible link specially made to break when the temperature reaches a pre-determined point. Normally the link should made to operate at 71oC (160oF) as B.S.S. 799 Part 5 1975, but other temperature should be catered for when required.

2.10.6 One heavy brass wood-screw threaded eye to secure the far end of the cable to the wall or ceiling.

2.10.7 One engraved warning notice to hang on the cable to prevent people walking into it.

PART 3 EXECUTION

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End of Section 22 05 23